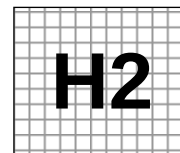


Civics Group	Index Number	Name (use BLOCK LETTERS)
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ST ANDREW'S JUNIOR COLLEGE
2013 JC2 Preliminary Examinations

H2 BIOLOGY

9648/3

Paper 3: Applications and Planning Question
(Mark Scheme)

Wednesday

18 September 2013

2 hours

Additional Materials: Answer Paper
 Cover Sheet for Section B

READ THESE INSTRUCTIONS FIRST

Write your civics group, index number and name on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagram, graph or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination,

1. Attach Question 5 to the **cover sheet** provided.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Paper 3	
1	/14
2	/14
3	/12
5	/20
Total	/60
4 (Planning)	/12

This document consists of **15** printed pages.

[Turn over

Answer **all** questions.

1. A student researcher attempted to clone the Human Growth Hormone (HGH) gene. Two separate samples of HGH DNA were obtained from DNA libraries and subjected to agarose gel electrophoresis. Results were shown in Fig.1.1.

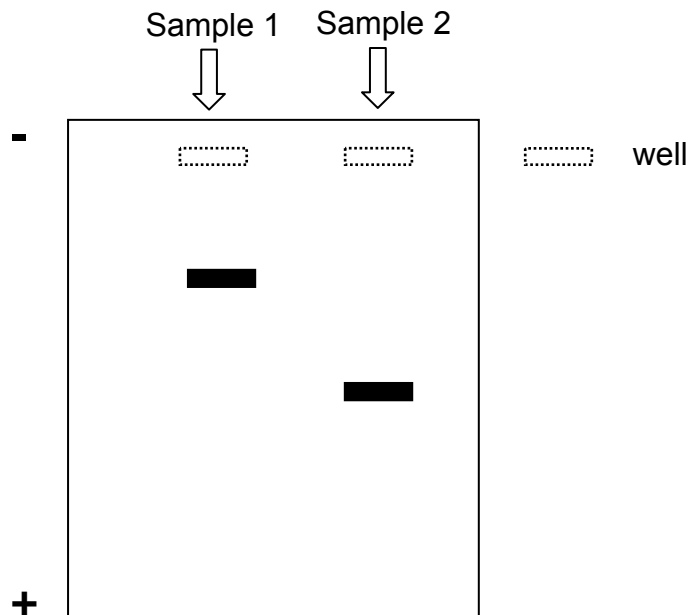


Fig. 1.1

(a)(i) Describe how the DNA fragments were separated into visible bands.

-[2]
- 1 negatively-charged DNA fragments migrate to the positively-charged electrode;
 - 2 **through** an agarose gel/matrix;
 - 3 DNA fragments separated by size / molecular weights
/ shorter fragments move faster than longer ones;
 - 4 invisible DNA bands revealed under UV light by staining the gel with a dye such as ethidium bromide;

Examiners comments:

Many candidates failed to read the question properly and missed out the EtBr staining followed by visualization under UV light in order to obtain “visible bands”

Subsequent cloning procedures involving *Escherichia coli* host cells yield functional HGH protein when Sample 2 was used, but not for Sample 1.

(ii) Based on the difference in the bands observed in Fig. 1.1, account for the failure to produce functional HGH protein for Sample 1.

-[2]
- 1 sample 1 has a higher molecular weight/larger size compared to sample 2;
 - 2 due to presence of introns in sample 1;
 - 3 bacterial host cells do not have RNA splicing machinery;
 - 4 ref. HGH gene not expressed correctly;

(iii) State the type of DNA library where Sample 2 was obtained from.

-[½]
- 1 cDNA library;

The student researcher has designed the following plasmid (Fig. 1.2) for use in her preliminary trials of a cloning experiment.

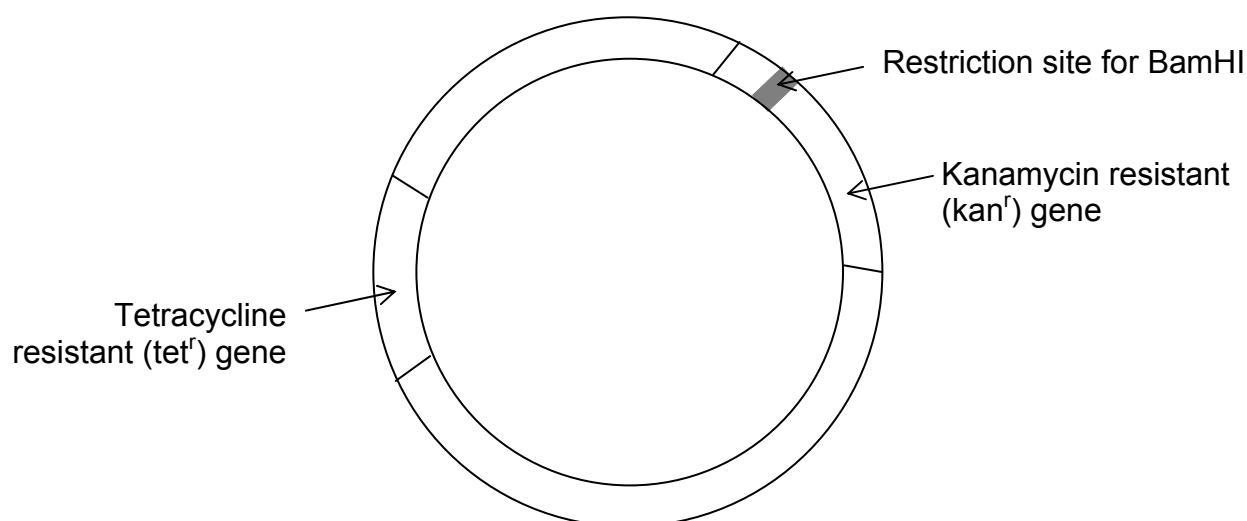


Fig. 1.2

(b) With reference to the properties of bacterial plasmids, provide 2 reasons why this plasmid in Fig 1.2 will not work well as a cloning vector.

.....[2]

- 1 no **origin of DNA replication** ;
- 2 no independent replication of the plasmid (and target gene)
/ cannot result in multiple copies of the plasmid (and target gene) within one bacterium;
- 3 no **prokaryotic** promoter;
- 4 which allows recognition by host cell's RNA polymerase (which is also prokaryotic)
/ no transcription/ expression of genes;

(c) The student researcher has since made improvements to the plasmid in Fig. 1.2 such that it is now a functional cloning vector.

Steps were then taken to create a recombinant plasmid using BamHI and the improved plasmid.

Restriction enzyme	Specific recognition site
BamHI	5'-G [^] A T C C-3' 3'-C C T A G [^] -5'

^ indicates where the restriction enzyme cuts

(i) Describe the natural function of BamHI restriction enzyme.

.....[1]

- 1 **protect bacteria** from attack by viruses/bacteriophages ;
- 2 cleave any foreign DNA that enters bacterial cell
/ by cutting up intruding DNA from other organism

(ii) Explain the main problem with using Sample 2 and BamHI in the formation of a recombinant plasmid and suggest how this problem can be solved.

.....[1½]

- 1 cDNA has blunt ends;
- 2 ref. incompatibility with sticky ends of plasmid after BamHI cuts;
- 3 ligate linker DNA;
- 4 to both ends of cDNA fragment;

(iii) Describe the subsequent steps in the formation of recombinant plasmid.

.....[2]

- 1 Use the same restriction enzyme to digest HGH cDNA and plasmid to generate complementary sticky ends
- 2 The plasmid and the HGH cDNA are **mixed** together
- 3 the sticky ends will **anneal** by complementary base pairing
- 4 DNA ligase added to seal the nicks by catalyzing the formation of phosphodiester bonds (between adjacent nucleotides on the sugar phosphate backbone)

(iv) With reference to Fig. 1.2, explain how you would identify transformed bacterial cells that have taken up the recombinant plasmid with the HGH gene using antibiotic selection.

.....[3]

- 1 grow bacteria on LB/tet plate ;
- 2 transformed cells can survive;
- 3 as they contain plasmid with intact tetracycline resistance gene;
- 4 **transfer** a small amount of each colony to an identified spot on LB/kan plate / **replica plate** on LB/kan plate;
- 5 bacterial colonies that cannot survive on LB/kan plates have the recombinant plasmid;
- 6 insertion of human gene disrupts sequence of kanamycin resistance gene;

Examiners comments

REJECT "recombinant plasmid" cannot survive. The correct phrasing should be "bacterial colonies containing the recombinant plasmid" cannot survive.

[Q1 Total: 14]

2. Severe combined immunodeficiency (SCID) is a genetic disorder characterized by lack of functional lymphocytes, resulting in defects in both T and B cells responses of the immune system. Patients are susceptible to opportunistic infections while children usually die if not treated.

Fig. 2.1 shows the inheritance of SCID in a family in which both parents are carriers. Only one of the sons suffer from SCID.

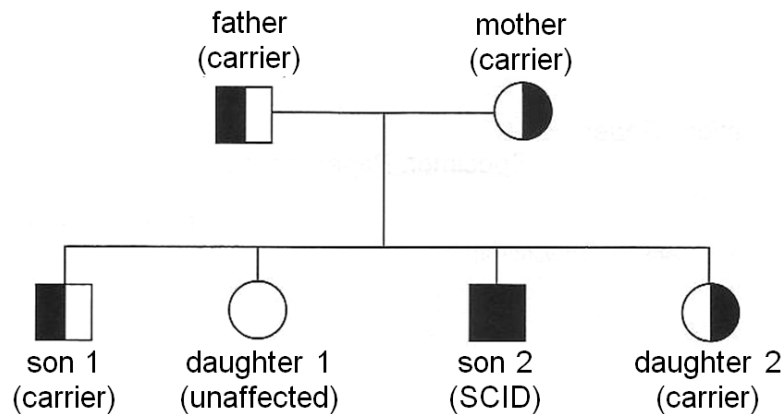


Fig. 2.1

(a) State the form of SCID that is being inherited in the family and the mode of inheritance.

.....[1]

1 ADA-SCID

2 autosomal recessive

(b) Two RFLP morphs have been found tightly linked to the adenosine deaminase (ADA) gene which can be used for genetic screening. A radioactive probe has been designed to reveal the two RFLP morphs.

Both RFLP morphs and radioactive probe are shown in Fig. 2.1 with the arrows representing the *PvuII* restriction sites. The asterisk (*) indicates the position of a mutation at the restriction site that results in the loss of the restriction site. Morph 2 was found to be linked to the mutant ADA gene.

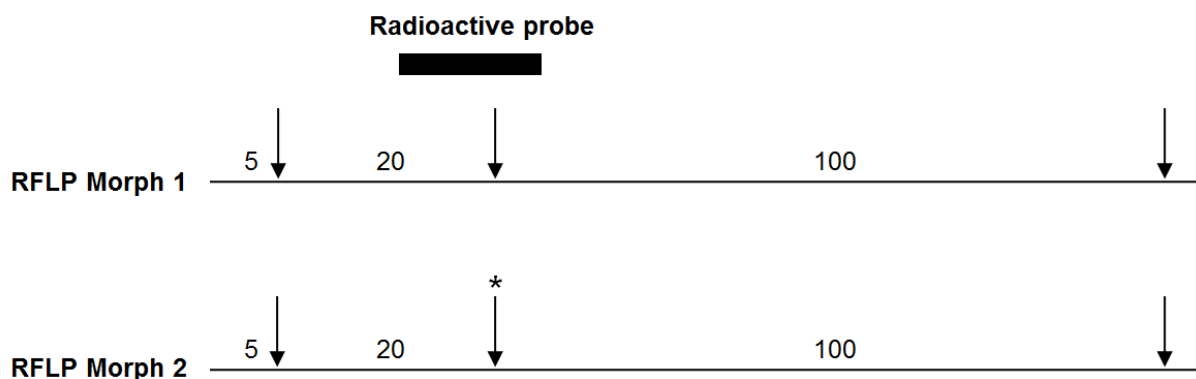
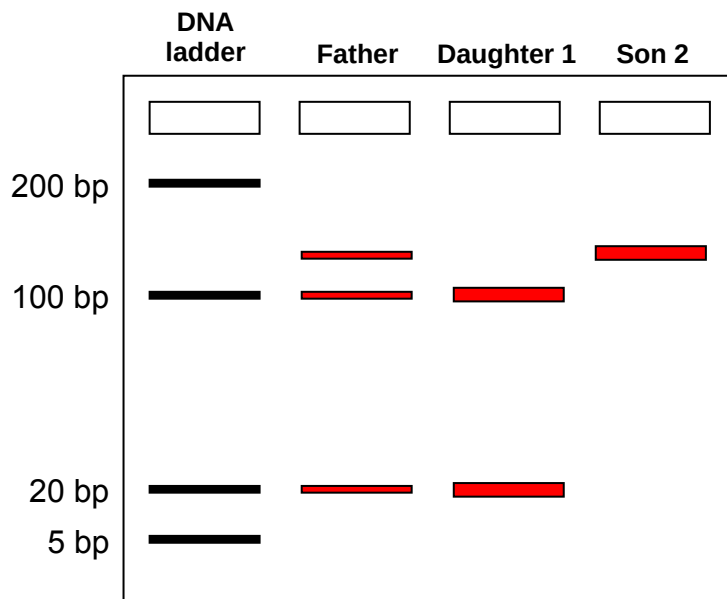


Fig. 2.2

(i) With reference to Fig 2.1 and Fig. 2.2, draw the resulting autoradiogram you would expect to see in the space below. [3]



For each individual,

- 1 Correct bands
- 2 Correct thickness

(ii) Explain the role of the DNA ladder. [1]

- 1 as a reference/standard / providing known DNA fragments of specific lengths
- 2 to estimate the size of unknown DNA molecules

(c) Apart from disease detection, other applications of RFLP include genetic fingerprinting and in genomic mapping.

(i) Explain why genes especially protein-encoding genes are seldom used as markers for genetic fingerprinting. [2]

- 1 sequence of protein-encoding genes are conserved
- 2 due to its important biological function;
- 3 mutation rate lower than non-coding region
- 4 not able to differentiate between individuals;

(ii) Explain the use of genetic markers in genomic mapping in terms of linkage mapping. [3]

- 1 Genetic markers are scattered throughout each of the chromosomes
- 2 used to determine the order/position of genes
- 3 Frequency with which two genetic markers are inherited together
- 4 is a measure of the closeness of the two loci on a chromosome
- 5 Recombination frequency / frequency of crossing over between 2 genetic markers
- 6 indicates the distance between two loci on a chromosome in map units / centiMorgan;

(d) Another method to detect ADA-SCID is through the use of PCR. Fig 2.3 shows the number of DNA molecules made using PCR, starting with one molecule.

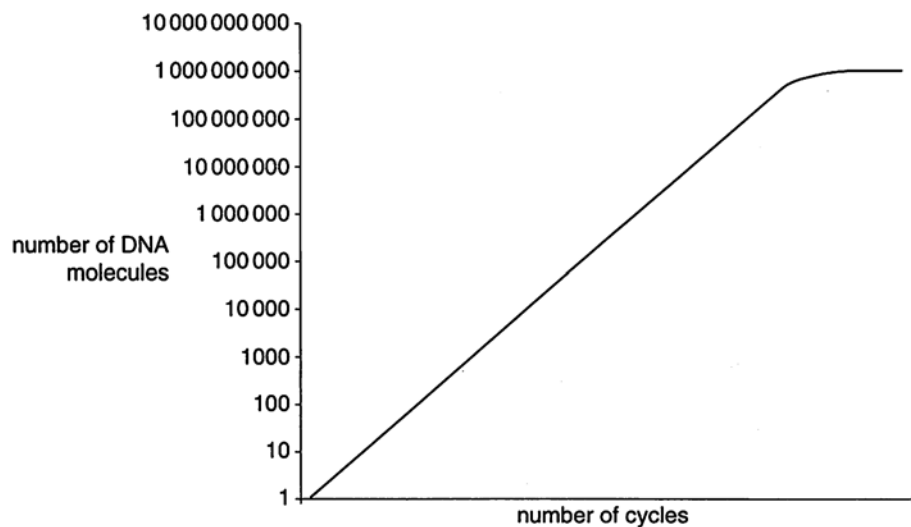


Fig 2.3

(i) Explain three advantages of PCR.

.....[3]

- 1 Millions of copies of target DNA can be obtained in a relatively short amount of time
- 2 Target DNA is doubled after every round of PCR / exponential increase / 2^n
- 3 Easy to set up a PCR reaction and the use of a thermocycler / PCR machine
- 4 availability of computer software for primer design / commercial synthesis of primers
- 5 Sensitivity of PCR
- 6 capable of amplifying sequences from minute amounts of target DNA
- 7 Specificity in amplifying only target sequences
- 8 as primers hydrogen-bond only to complementary sequences
- 9 A broad range of nucleic acid sources are suitable templates for PCR amplification / Robustness
- 10 permit amplification of specific sequences from material in which the DNA is badly degraded or embedded in a medium
- 11 Cell-free method of DNA replication
- 12 requires no cleanup of unwanted cellular debris or vector DNA

(ii) Suggest two reasons for the leveling off of the graph from cycles 17 to 20.

.....[1]

- 1 Nucleotides are used up (hence is unable to make complementary chains)
- 2 Primers are used up
- 3 the template DNA strands anneal with each other instead of with the primer
- 4 *Taq* DNA polymerase gradually becomes denatured ref. no elongation

[Q2 Total: 14]

3.

(a) Many varieties of rice, *Oryza sativa*, die quickly when totally submerged by flood water during the monsoon season. Typically, submergence-intolerant plants respond to submerged conditions by elongating rapidly.

In 2006, a gene, *Sub1A*, on rice chromosome 9 was found to be involved in tolerance of prolonged submergence.

An allele, *Sub1A-1*, was found only in submergence-tolerant rice. Varieties that are submergence-intolerant either have a second allele, *Sub1A-2*, or lack the gene altogether.

(i) *Sub1A-1* codes for a protein that controls the transcription of the gene for alcohol dehydrogenase. This gene is activated when the plant is in anaerobic conditions.

Explain the importance to submerged rice plants of producing alcohol dehydrogenase.

.....[2]

- 1 Lack of oxygen in submerged conditions
- 2 leads to anaerobic respiration
- 3 Alcohol dehydrogenase catalyse conversion of ethanal to ethanol
/ regeneration of NAD^+
/ hydrogen atoms from NADH donated to ethanol to form NAD^+
- 4 allows for generation of **ATP** (by glycolysis)

Examiners comments

In plants (i.e. alcoholic fermentation), alcohol dehydrogenase enzyme catalyses the reduction of ethanol/acetaldehyde to produce ethanol, with NAD^+ being regenerated in the process. In animals, alcohol dehydrogenase is used in the liver in an opposite reaction to breakdown ethanol..

A variety of submergence-intolerant rice, which lacked the gene *Sub1A*, was genetically engineered to express *Sub1A-1*.

The mean heights of three types of rice plants were compared before and after 10 days of submergence in water. The types of rice plants were:

I : submergence-intolerant plants

I⁺ : submergence-intolerant plants genetically engineered to express *Sub1A-1*

T : submergence-tolerant plants

The results are shown in Fig. 3.1.

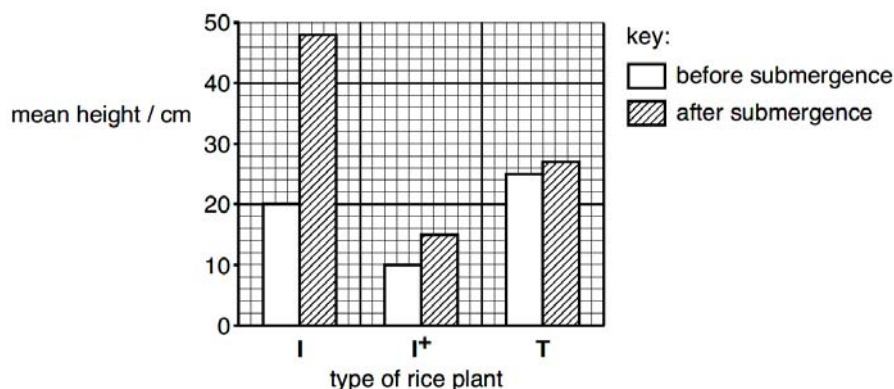


Fig. 3.1

(ii) With reference to Figure 3.1, compare the effect of submergence on the elongation of the three types of rice plants.

.....[2]

- 1 All three plants elongate during submergence ;
- 2 ref. comparison of elongation of I, I⁺ and T plants
/ eg. Expression of Sub1-A1 in I⁺ and T resulted in reduced elongation ;
- 3 I – 28 cm elongation / 2.4 times elongation
- 4 I⁺ – 5 cm elongation / 1.5 times elongation
- 5 T – 2 cm elongation / 1.08 times elongation

The allele *Sub1A-1* could be introduced into high-yielding variety of submergence-intolerant rice either by genetic engineering or by selective breeding.

(iii) Suggest two advantages of using genetic engineering, rather than selective breeding, to introduce the allele.

.....[1]

- 1 More specific in transfer of allele conferring high yield
- 2 Does not dilute desirable traits of high yield variety
- 3 Genes from different species can be combined (to produce a transgenic organism)
- 4 Shorter time required to see results compared to traditional selective breeding which may take several generations

(b) Round-Up ReadyTM soybeans are genetically modified to be resistant towards Round-UpTM herbicide, a glyphosate-based herbicide that normally inhibits an important enzyme in the aromatic amino acid synthesis pathway in plants. This allows farmers to freely apply Round-UpTM to destroy weeds without affecting the Round-Up ReadyTM soybeans.

Round-Up ReadyTM soybeans may be produced by first transfecting soybean callus cells with the resistance gene from *Agrobacterium*.

(i) To obtain a callus culture, suitable explants must first be chosen. Suggest a suitable explant and explain your choice.

.....[1]

- 1 eg. apical or axillary buds / shoot tips / root tips
- 2 contain meristematic tissues which are (actively dividing, undifferentiated and) **disease-free**

(ii) Explain three advantages of growing soyabean plants from tissue culture rather than from seeds.

.....[3]

- 1 **genetically identical**
- 2 possess the **desirable features of the stock plants**
- 3 allows the **rapid** multiplication of plants
- 4 giving rise to **bulk production** of genetically identical plants
- 5 new plants are **disease-free**;
- 6 leading to higher **yield** / better **quality**
- 7 **genetic modifications** possible (with help of protoplast cultures)
- 8 **desirable traits** can be introduced
- 9 Take up **little space** when compared with plants growing in fields
- 10 can be **grown intensively**
- 11 Plantlets are light and small in **size**
- 12 can be air-freighted and **transported** easily/cheaply/in large quantities, increasing international trade
- 13 **Independent of climate** changes so plants can be produced continuously/at any time of year
- 14 **flexibility** in meeting consumer demand
- 15 possible to **standardise** the conditions for growth and obtain many batches of identical plants
- 16 ensures product **uniformity**

(iii) Soyabean plants can reproduce sexually by means of seeds. Soyabean plants grown from seeds are very variable in their yield. Explain why.

.....[2]

- 1 there is genetic variation due to

During gamete formation, there is

- 2 **Crossing over**
- 3 between non-sister chromatids of homologous chromosomes during prophase I
- 4 **Independent assortment** of homologous chromosomes
- 5 Seeds formed through **random fusion** of gametes

Genetically modifying soybean plants to be resistant towards Round-Up™ herbicide might threaten the environment and human safety.

(iv) State one possible adverse effect on:

The environment

-[½]
- 1 Seeds from GM crops might be carried to other places and establish themselves as weeds
 - 2 Cross-pollination between the GM crops and their wild relatives may spread the resistance to weeds
/ Emergence of vigorous weeds with herbicide-resistant genes

Examiners comments

Some candidates were confused between herbicide-resistant plants and pesticide-resistant plants.

Human safety

-[½]
- 1 Introduction of foreign gene(s) may result in production of secondary **metabolites** which may be toxic to animals themselves and/or livestock/humans that consume them.
 - 2 New **proteins** in GM plants may be potentially allergenic to humans that consume them.
 - 3 Vectors used in introducing the herbicide resistant gene can contain genes for antibiotic resistance which may pass from the plant to the *E. coli* in the gut when the soybean is eaten, making the bacteria resistant to antibiotics

[Q3 Total: 12]

4. Planning question

The goshawk is an endangered species protected by the Convention of International Trade in Endangered Species of Wild Fauna and Flora (CITES).



The Royal Society for the Protection of Birds (RSPB) has found a man in possession of four young goshawks and an adult female. He claimed to have bred the young birds using his adult female and a male bird borrowed from another keeper of these birds of prey. The RSPB suspects that the young goshawks were poached from the wild and not bred in captivity as claimed by the man.

Plan an investigation using DNA fingerprinting to verify if the young goshawks were bred in captivity or poached from the wild.

[Total: 12]

Your planning must be based on the assumption that you have been provided with the following equipment and materials

- tissue samples (feather tips) from the four young goshawks, the adult female and the adult male under investigation
- pestle and mortar
- DNA extraction buffer solution
- ice-cold ethanol
- glass rods
- microcentrifuge tubes
- centrifuge
- restriction enzyme
- PCR reagents and equipment
- agarose gel electrophoresis reagents and equipment
- suitable source of electrical current
- radioactive probe
- nitrocellulose membrane
- autoradiography equipment

Your plan should have a clear and helpful structure to include

- an explanation of the theory to support your practical procedure

- a description of the method used, including the scientific reasoning behind the method
- the type of data generated by the experiment
- how the results will be analysed including how the origin of the organism can be determined.
- relevant risks and precautions taken

Theoretical consideration or rationale of the plan to justify the practical procedure (1 mark)

- 1 RFLP refers to differences in the restriction sites on homologous chromosomes that leads to different restriction fragment patterns.
- 2 DNA fingerprints makes use of polymorphisms that consist of repeated short tandem repeats (STRs).

Method of DNA extraction including homogenization and use of buffers (2 marks)

- 1 Homogenise the feather tips of a goshawk by grinding using the **pestle and mortar** in the presence of **DNA extraction buffer solution**.
- 2 **Centrifuge** homogenised mixture to separate DNA from the rest of the cell debris. Use a micropipette to transfer the **supernatant** which contains the DNA into another clean microcentrifuge tube
- 3 Add **ice-cold ethanol** to precipitate the DNA out of solution. Using the **glass rod** to spool up the precipitated DNA and transfer it to a fresh microcentrifuge tube.
- 4 Resuspend the DNA with restriction enzyme buffer.
- 5 Repeat steps the preparation of DNA from the **other goshawk tissue samples** and. ensure that the same mass of tissue is used to ensure an equivalent amount of DNA can be produced for analysis.

Selection of restriction enzyme and reason for the selection (1 mark)

- 1 Add 5 µl of EcoRI to sample. Incubate enzyme with DNA samples for **2 hours at 37°C**.
- 2 EcoRI will cleave the DNA segments at different sites, generating different sized fragments for different individual goshawks.

Amplification of DNA fragment using PCR including detail of PCR (2 marks)

- 1 The DNA is added in a PCR tube with Taq polymerase, free deoxyribonucleotides and primers designed to flank the DNA sequence to be amplified.
- 2 The mixture is heated to **95°C** to break the hydrogen bonds so that the double-stranded DNA separates into single strands.
- 3 The mixture is then cooled to **55°C** to enable the annealing of DNA primers to complementary sequences flanking the target sequence to be amplified
- 4 The mixture is heated to **72°C** for Taq polymerase to catalyse the synthesis of a complementary strand of DNA for each of the single strands of the target DNA.
- 5 PCR is needed to **generate sufficient** DNA for genetic fingerprinting or analysis / **Repeat cycles for 30 times**. The amount of DNA produced will double each time, resulting in amplification of target DNA sequence.

Separation of fragments by agarose gel electrophoresis and the principles behind the separation (2 marks)

- 1 Prepare agarose gel cast and set up electrophoresis chamber.
- 2 Add 5 µl loading dye to DNA samples. Spin the samples down using centrifuge.
- 3 Load 10 µl of DNA ladder into the first slot. Load 10 µl of individual DNA samples into the subsequent slots.
- 4 Conduct gel electrophoresis at 100V till tracking dye move to $\frac{3}{4}$ length of gel.
- 5 DNA is negatively-charged and will migrate to the positive electrode. Smaller fragments move through the gel faster than larger fragments.

Transfer of DNA onto nitrocellulose membrane (3 marks)

- 1 The gel is treated with sodium hydroxide to cause the double-stranded DNA to denature, separating it into single strands. Denaturation is necessary so that the DNA will stick to the membrane and be hybridized by the single stranded DNA probe.
- 2 A sheet of nitrocellulose membrane is placed on top of the gel. Pressure is applied evenly to the gel by placing a stack of paper towels and a weight on top of the membrane and gel. This causes the DNA to move from the gel onto the nitrocellulose membrane by capillary action, where it sticks.
- 3 The membrane is then baked to permanently crosslink the DNA to the membrane.

Hybridisation with radioactive labeled DNA probe

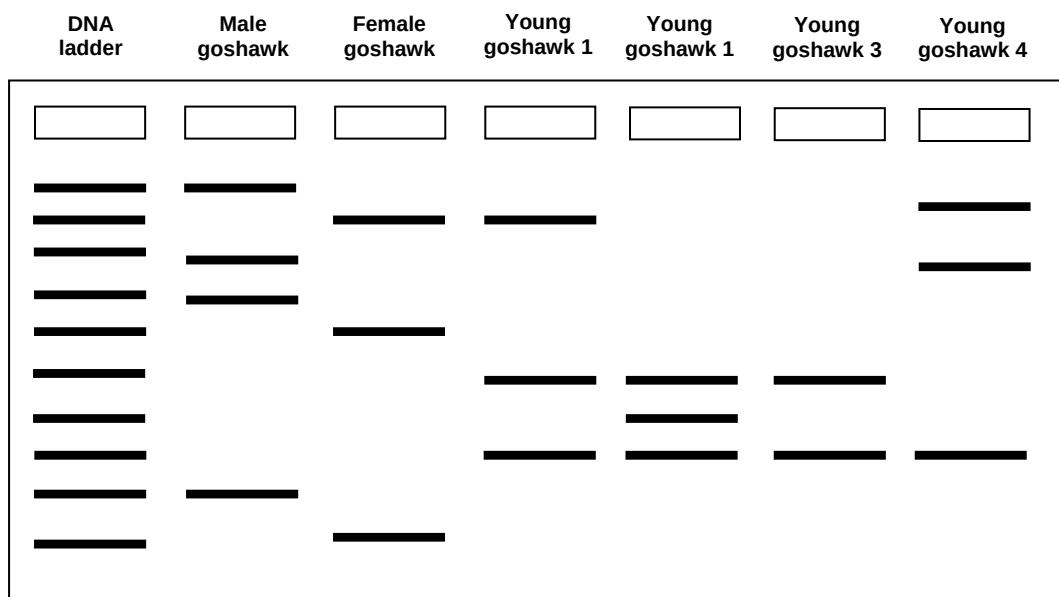
- 4 The membrane is treated with a single-stranded radioactive hybridization probe which is complementary to the target STR sequence.

Autoradiography method and method of band visualization

- 5 After hybridization, excess probe is washed from the membrane
- 6 The pattern of hybridization is visualized on X-ray film by autoradiography.

Significance of matching bands and proposed method of result feedback to RSPB (1½ mark)

- 1 If the young goshawks are bred in captivity, all bands in the offspring can be assigned as coming from either the mother or the father.
- 2 If the young goshawks are poached from the wild, there will be unassigned bands.
- 3 Autoradiograph diagram



Risks and safety measures (1 mark)

- 1 Radioactive probe is carcinogenic. Use isolated area for radioactive probe hybridisation, work behind a radioactive shield and use latex gloves when handling samples.
- 2 To prevent electrical shock, ensure hands are dry before using the electrical appliance such as the electrical power supply of agarose gel electrophoresis. Clean all spillage before running the gel to ensure all surfaces near the electrophoresis gel chamber are dry.

Control (1 mark)

- 1 Extract DNA from random goshawks and conduct the same DNA fingerprinting experiment. Unrelated goshawks should have unassigned bands compared to the male and female parents.
- 2 Otherwise, new probes for other regions of the genome should be designed to differentiate between individuals.

Reliability (½ mark)

- 1 ref. At least 3 replicates and at least 3 repeats

1 Free-response question

Write your answer to this question on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labeled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

5 (a) Compare the properties, features and functions of zygotic and embryonic stem cells. [5]

(b) Using one named example of an adult stem cell, describe its normal role and how it could be used in gene therapy against SCID using a retrovirus. [8]

(c) Discuss the social and ethical considerations for the use of gene therapy. [7]

[Total: 20]

(a) Compare the properties, features and functions of zygotic and embryonic stem cells. [5]

Similarities (max 1)

- 1 Self-renewal ability / divide continuously by mitosis to produce new stem cells
- 2 Unspecialized / do not have cell-specific structures to carry out specific functions
- 3 Undifferentiated

Differences

Zygotic stem cells	Embryonic stem cells
Isolated from morula / zygote;	Isolated from inner cell mass of blastocyst;
Totipotent, pluripotent and multipotent;	Pluripotent, not totipotent but are multipotent;
able to differentiate into any cell type to form any organ or type of cells, including extra-embryonic membranes ;	able to differentiate into almost any cell type to form any organ or type of cells, except those of the extra-embryonic membranes;
ability to differentiate into any cell type to form whole organisms ;	give rise to various organs in organism / multiple specialized cell types that make up the heart, lung, skin and other tissues in the developing foetus

(b) Using one named example of an adult stem cell, describe its normal role and how it could be used in gene therapy against SCID using a retrovirus. [8]

- 1 Named example: blood stem cells;
- 2 Differentiate into cells of the same lineage;
- 3 E.g. red blood cells, white blood cells and platelets;
- 4 Replace cells worn out from normal wear and tear;
- 5 (blood stem cells) isolated from bone marrow (of patient);
- 6 No cell rejection;
- 7 Cultured *in vitro* first to **increase numbers**;
- 8 Attenuate retroviral vector by removing viral genes that cause harm (eg. lysis);
- 9 mRNA of target gene inserted in vector;
- 10 ref. ADA gene / IL2RG gene;
- 11 Recombinant vector allowed to infect cultured blood stem cells;
- 12 Viral genome is reverse transcribed;
- 13 and resulting cDNA is then integrated into host cell DNA;
- 14 can be **stably propagated** by chromosomal replication following cell division
- 15 Selection of transformed blood stem cells on selection plates;
- 16 **Transformed** blood stem cells grown in culture to **increase number** of cells
[e.c.f from pt 7]
- 17 Transgenic blood stem cells grown reimplanted/injected into patient;
- 18 ref. normal functional protein product, thereby restoring the correct function of the target cells and altering the phenotype
- 19 functional **T and B lymphocytes** in the patient;
- 20 to fight off infections / development of functional immune system;

(c) Discuss the social and ethical considerations for the use of gene therapy.[7]

Benefit of gene therapy

- 1 Potential for tremendous **patient benefit**
- 2 **Germline therapy** can eradicate disease from subsequent generation and prevent transmission of serious genetic diseases from parent to offspring

Interfere with evolution

- 3 elimination of unwanted alleles from the gene pool will **decrease genetic variation**
- 4 genes that are damaging under some conditions may be advantageous under other conditions
/ germ-line gene therapy would forever change the genetic makeup of an individual's descendants / the human gene pool would be permanently affected
/ detrimental to the survival of our species in future

Safety

- 5 question of **safety and efficiency** of treatment and therefore unknown long term consequences to the well-being of the recipients.
- 6 raises questions over who should participate in a **human trial** – the terminally ill with no treatment options or anyone who understands the potential risk but is a willing patient

Cost

- 7 gene therapy is currently very expensive, require expertise and equipment found only in major medical centres, concerns about gene therapy become a luxury **available only to the rich** and powerful.
- 8 possible genetic **enhancements creating an advantage** for those who can afford the treatment

Eugenics social policies

- 9 type of circumstances that determines whether genomes should be altered / what is normal or a disability or disorder, and who decides / potential for **non-therapeutic enhancement** possibilities / eugenic social policies
- 10 concerns about the widespread use of gene therapy making society **less accepting** of people who have mild disorders / genetic diseases or less able / lead to new definitions of “normal (e.g. intellect / height / strength etc.)

Privacy

- 11 concerns about the protection of **privacy** and confidentiality of medical information of patients involved in clinical trials
- 12 which may have implications on (medical) **insurance coverage / employability**

Interfering with life

- 13 opposition to gene therapy based on **religious grounds**, believing that altering genetic material is against God’s will
- 14 the **unborn child** who would be affected by germline gene therapy cannot choose whether to have the treatment